

Preprocessing Step

- Missing Values
 - References & Authors: Empty List
 - Abstract: Empty String
- Normalize
 - Lowercase
 - Normalize whitespace
- Cleaning special characters
- Vectorization (TF-IDF)

Pre-popularity EDA Step

- Univariate findings:
 - Box Plots, Scatter Plots, Outlier removals, Histograms
- Bivariate & Trivariate Findings
 - Focus on interesting relationships and test our assumptions
 - 2D and 3D Scatter Plots
- Variance Findings
 - PCA's (2 components, 3 components)

Clustering Step

- Ordering by certain features
- K-Means
- Hierarchical methods (Dendrogram)
- Density-based clustering w/ DBSCAN
- Iterate over 3 models with different hyperparameters, looking for the best clustering
- Creating class labels using natural forming clusters
- Detect any particular anomalies in the dataset
- Identify what makes a paper popular, and draw some conclusions
- Maybe do some extra crazy ish (Semi-supervised learning)

Classification Step

- Use topic labels produced by clustering as class labels, then train models to learn whether title and abstract features can reliably predict a paper's topic groups
- Use Decision Trees, Random Forest, and Multinomial Naive Bayes
- Evaluate metrics using Macro-F1, Micro-F1, Accuracy, Precision, Recall, and training time
- Testing assumptions:
- Adding venue information to title and abstract features will improve topic classification performance
- Topic clusters with more distinct vocabulary will be easier to classify compared to others
- Some topic clusters will have a higher average number of citations which suggest what areas have had the greatest influence